CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Endomorphisms of hyperbolic groups

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Abstract

The Bounded Reduction Property (or Bounded Cancellation Lemma) was introduced by Cooper for automorphisms of free groups and has played an important role in the dynamical study of endomorphisms of (virtually) free groups since then. In this talk, we will present some (equivalent) geometric versions of this property for endomorphisms of hyperbolic groups and show how we can use it to obtain an affirmative answer to a question from Araújo and Silva as to whether every nontrivial uniformly continuous endomorphism of a hyperbolic group with respect to a visual metric satisfies a Hölder condition. This, together with a result from Paulin, shows that all endomorphisms admitting a continuous extension to the Gromov completion have finitely generated fixed subgroup. In the end, some further work on this topic will be discussed.